

Design of Lightweight Aerial Manipulator With a CoG Compensation Mechanism

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Abstract — This paper discusses the design and validation processes involved in the development of a lightweight aerial manipulator. The structure consists of 3D printed links with a counterweight on one end and a gripper on the other, its task is a pick and place operation. The manipulator is inspired from pantograph mechanism, which acts as a center of gravity (CoG) compensation active mechanism. This mechanism is intended to be attached beneath a drone or quadrotor to solve the CoG shifting problem during manipulator operation. An experimental test bench is developed to simulate real flying operations. The mechanism base tilting pitch angle during movement is obtained and compared both with and without the using of proposed aerial mechanism to validate the performance and stability of the quadrotor while used for pick and place applications.

Keywords— Aerial manipulator, Center of Gravity (CoG) balancing, Unmanned Aerial Vehicle (UAV).

I. INTRODUCTION

Drones or unmanned aerial vehicle (UAV) are becoming a more integral part of all societies nowadays. Traditionally, a drone is designed to fly using an interface between the user and itself, and they sometimes have limited surveillance devices mounted on them like cameras due to payload and stability issues. However, several attempts were addressed by research to deploy manipulators on board these drones, the objective of a drone manipulator or aerial manipulators is to extend the applications of normal drones and allow them to perform a task which requires action without any hands-on interaction [1,5, 9- 11]. Operations such as pick and place, crop spraying, perform precarious maintenance or inspection tasks can be carried out using this method [2,5]. The latter could be especially useful in dangerous situations, where a worker must do something at a high altitude, such as: on wind turbines, high voltage line inspection, collecting samples and inspection of water dams, volcanoes, forests and lakes. All these applications provides a challenge for the drone and the attached manipulator to accomplish, since the center of gravity (CoG) of the drone-manipulator system shifts when an object is gripped by the end effector, especially with large workspace for the manipulator, causing the drone to become aerodynamically unstable and potentially crash.

In literature, different mechanisms have been proposed to rectify the CoG shifting problem; each has its own advantages and drawbacks. These mechanisms can be categorized as: classical serial mechanism, parallel mechanism, manipulator attached to the drone without any CoG balancing, or mechanisms with CoG balancing. Among the latest, a sliding object (or batteries) mechanism [3], the proposed solution discussing this mechanism considers the UAV and the aerial manipulator (with a 6 DOF arm) as two separate independent systems [3]. This control system is divided into three layers, a sliding mechanism moves the counterweight (batteries) in the opposing direction of the

arm, propellers compensate for residual static momentum induced by the manipulator's gravity on the multirotor, and an estimator of external generalized forces acting on the multirotor is implemented to compensate everything else has not been taken into account in the previous layers. An onboard sliding mechanism [4], utilizes a link pinned to the body of the drone to allow for compensation in the shift of the CoG of the system [4]. The manipulator itself is a fixed part to the end effector, which can extend by means of a hydraulic cylinder. The battery link's connection and the end effector contribute to two degrees of freedom for the system. The downside is that since the link is always moving and it is protruding outward from the base, the mechanism cannot be used for applications in tight spaces, since the link should be kept away from walls or any structure where collision is imminent. The pantograph mechanism [5], was tasked with preserving the quadrotor's stability using an algorithm that predicts the payload's mass variation which results from the catching or releasing of objects and thus moves the CoG mechanism accordingly using motors [5]. Moreover, the pantograph mechanism ensures that the resulting shift of the hosting platform's CoG does not exceed 0.5 mm for a specific payload mass at steady state. The research in [6] discusses the idea of using a symmetrical scissors mechanism in parallel aerial manipulators. The proposed solution uses three parallel manipulators in order to satisfy the stability of the drone. Each arm, made of a lightweight acrylic material, has 2 DOF. The total weight of the aircraft is kept under 950 g, and the aircraft can carry objects below 200 g. This paper propose a mechanism for CoG balancing, the active mechanism rectifies the issue of COG shifting by utilizing a pantograph like structure [5,12,13]. It consists of 3D lightweight printed links, a gripper on one end and an active counterweight link on the other.

II. BASIC DESIGN

A. Design Requirements

The proposed design shall consider multiple parameters pertaining to certain elements of the system. The mechanism should have an acceptable workspace, so objects do not have to be approached closely. The links should hence have a suitable length due to the created moment. The proposed design is tasked with maintaining in-flight stability while grasping an object at the end effector. The links should be able to withstand the stress created by the grasped object, in addition to providing support and maintaining the structural integrity of the mechanism, rigidity and stiffness are to be considered.

B. Conceptual Design

The proposed mechanism, as shown in figure 1, consists of four 3-D printed links; bearings, washers and 3-D printed pins used to integrate the mechanism. Each link has at least two holes, where a bearing will be inserted snugly. A washer is placed in contact between the bearings to ensure a uniform pressure distribution in the joint, which will be held in place

using one hollow and one normal pin. The 2-outer links, L_1 , have the same length, and the 2-inner links, L_2 and L_3 , provide support and help in conserving the mechanism's structural integrity. The links were chosen to be printed out of PLA plastic. The mechanism was designed to carry a payload of 200g. The payload is of utmost importance for understanding how much stress the links will undergo while the mechanism is in operation. This could be either indoors or outdoors. When the mechanism grasps an object, the center of gravity of the system shifts, so the servo motors have to move the link (L_1) with the counterweight to a certain extent to make sure no net moment exists about the vertical axis of the drone platform.

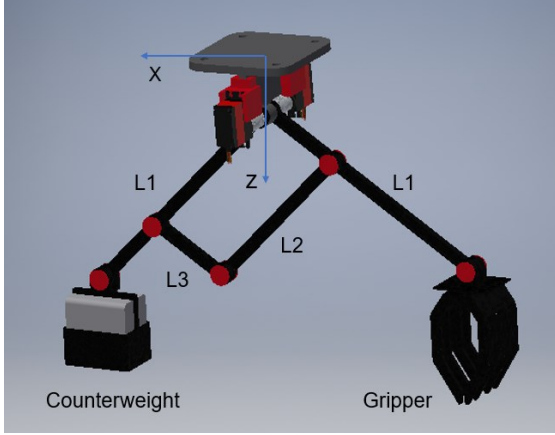


Fig. 1. Mechanism CAD model

Figure 2 shows the actual mechanism being tested during the testing phase. The mechanism consist of four main parts as follows: the yellow portion corresponds to the base which includes the microcontroller, sensors and motors, the green rectangle corresponds to the manipulator mechanism, the blue circle is for the counterweight and the red is the gripper.

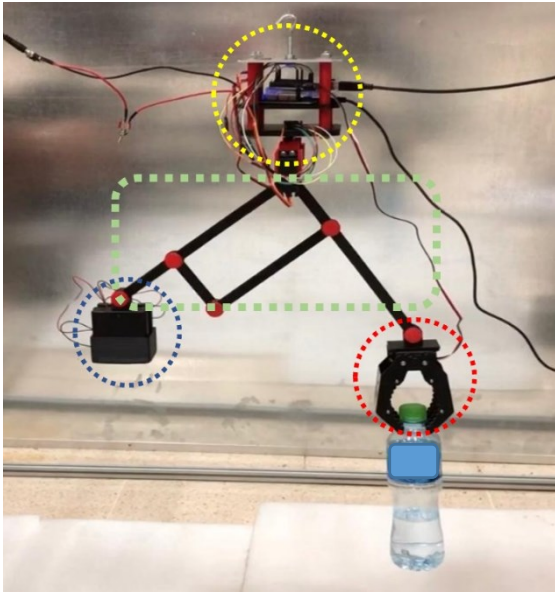


Fig. 2. Final manipulator prototype

III. MODELING AND ANALYSIS

A. Modeling and COG Analysis

As shown in figure 3, the left side of the manipulator is the side with the counterweight, and the right side contains the gripper. To analyze the center of gravity of a composite body involves two steps. Since the mechanism is assumed to be in one plane, the x-z plane, where the z axis is the axis pointing downward of the drone, two coordinates for the center of gravity exist. To find the x-coordinate of the CoG, the equation (1) for composite body center of gravities is applied, where n is the number of elements. The same is used for the CoG z-coordinate:

$$x_{total}m_{total} = \sum_{i=1}^n x_i m_i \quad (1)$$

All components with a notable mass (m_i) need to be considered; these include the links, the bearings, the washers, the normal and hollow pins, the counterweight and the end effector as shown in figure 3. Finally the z and x coordinates for CoG are given in equations 2 and 3, respectively.

$$Z_{total} = \frac{\left\{ \begin{aligned} & \left[\frac{L_1}{2} \times \sin(\theta_1) \times m_1 \right] + [L_1 \times \sin(\theta_1) \times (m_c + m_j)] + \dots \\ & \left[(L_2 + \frac{L_3}{2}) \times \sin(\theta_1) \times m_3 \right] + [(L_3 + \frac{L_2}{2}) \times \sin(\theta_2) \times m_2] + \dots \\ & \left[(\frac{L_1}{2}) \times \sin(\theta_2) \times m_1 \right] + [L_1 \times \sin(\theta_2) \times (m_e + m_j)] + \dots \\ & [L_2 \times \sin(\theta_1) \times m_j] + [(L_1 + L_3) \times \sin(\theta_1) \times m_j] + \dots \\ & [L_3 \times \sin(\theta_2) \times m_j] \end{aligned} \right\}}{m_{total}} \quad (2)$$

$$X_{total} = \frac{\left\{ \begin{aligned} & \left[\frac{L_1}{2} \times \cos(\theta_1) \times m_1 \right] + [(L_2 - \frac{L_3}{2}) \times \cos(\theta_1) \times m_3] + \dots \\ & [L_1 \times \cos(\theta_1) \times (m_c + m_j)] + [\frac{L_1}{2} \times \cos(\theta_2) \times m_1] - \dots \\ & [(L_3 - \frac{L_2}{2}) \times \cos(\theta_2) \times m_2] - [L_1 \times \cos(\theta_2) \times (m_e + m_j)] + \dots \\ & [L_2 \times \cos(\theta_1) \times m_j] + [(L_2 - L_3) \times \cos(\theta_1) \times m_j] - \dots \\ & [L_3 \times \cos(\theta_2) \times m_j] \end{aligned} \right\}}{m_{total}} \quad (3)$$

Where m_c is the mass of counterweight, m_e is the mass of end effector, and m_j is the mass of joint includes the mass of 2 bearings, 1 washer, 1 pin and 1 hollow pin. Table 1, shown below, provides all important dimensions for the mechanism to be used in calculating the CoG. As it can be noticed from equations 2 and 3, a CoG with a non-zero x-coordinate insinuates that a net moment exists about the drone platform pitch axis, which means that the system will tilt and lose control, rendering it unsafe [7]. The counterweight mechanism is intended to rectify this by motorizing the outer link (L_3) which allowed moving to compensate the shifting of the CoG of the mechanism.

Table 1: Mechanism Dimensions

Part Name	Length (mm)
Link 1 (L_1)	250
Link 2 (L_2)	170
Link 3 (L_3)	80

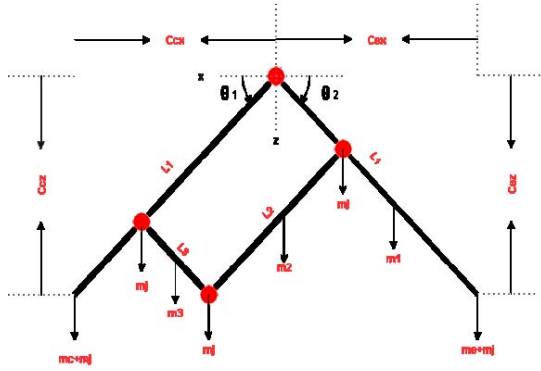


Fig. 3. Model of the manipulator mechanism

With reference to figures 2 and 3, the range of angles for θ_1 and θ_2 is $[0^\circ 85^\circ]$ due to physical constraints concerning the link thicknesses. The workspace of the mechanism is defined by the path line traced by the endpoints of L_2 , which is shaped like an arc as shown in figure 4 and rotated by the drone about the z axis.

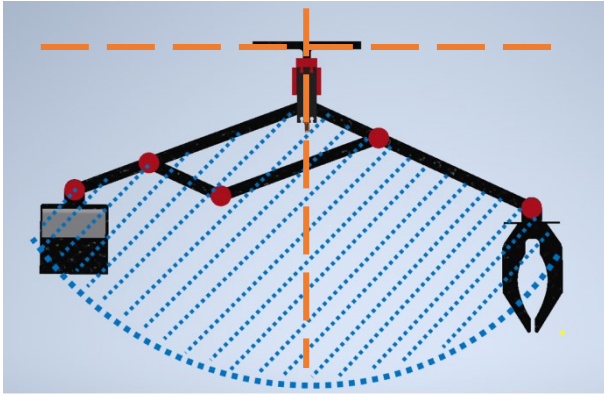


Fig. 4. Work space of the manipulator

B. Center of Gravity Balancing

To demonstrate the active CoG balancing, for a given end effector position, two different fixed angles for θ_2 , were selected to be 45° and 60° . θ_2 is defined from the negative x-axis downwards. Equations 2 and 3 are used to calculate the CoG coordinates for every value of θ_1 between 0 and 90° for the two cases, while the gripper carries a payload of 200 g. The resulted CoG x-coordinate vs. θ_1 was illustrated in figures 5 and 6. As it can be noticed from the plot in fig. 5, the θ_1 angle required to place the counterweight link such that the CoG of the mechanism is located on the z- axis of the drone platform is 49° . This point was found numerically using a manually synthesized bisection root solving method as shown in figure 7, using 0° as the lower limit and 90° as the upper limit, since the solution will inevitably fall between these two abscissas. For the second case, as shown in fig. 6, the θ_1 angle required to place the counterweight link such that the CoG of the mechanism is located on the z- axis of the drone platform about 20° .

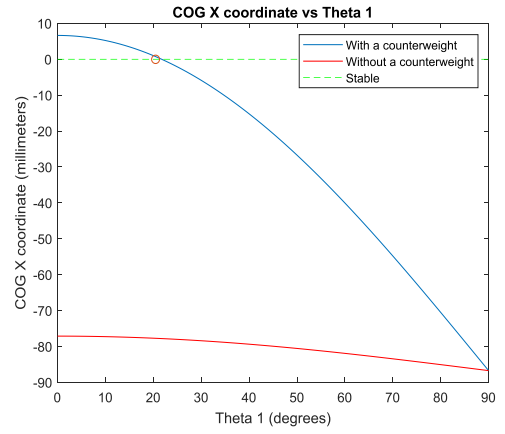


Fig. 5. CoG x-coordinate vs θ_1 ($\theta_2 = 45^\circ$)

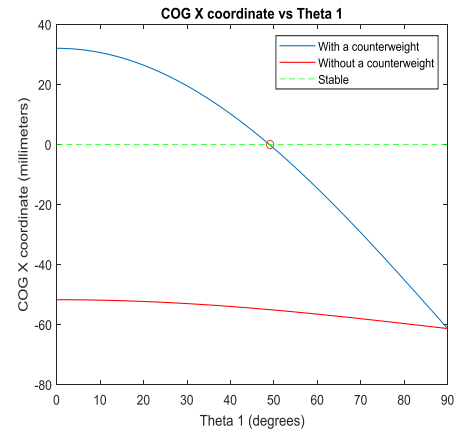


Fig. 6. CoG x-coordinate vs θ_1 ($\theta_2 = 60^\circ$)

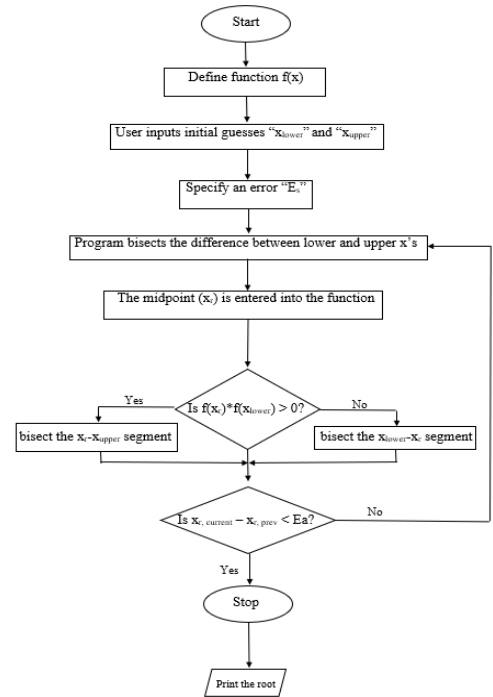


Fig. 7. Bisection method flow chart

The importance of proposed counterweight balancing mechanism is highlighted by plotting the CoG position without utilizing an active counterweight balancing part of the mechanism. As illustrated in figures 5 and 6, the importance of the proposed mechanism is demonstrated for such application, the x coordinate of the CoG in both cases is away from the equilibrium point, as shown in red in both plot.

IV. EXPERIMENTAL RESULTS

For experimental validation a real time independent joint control was implemented for the mechanism. An SSC-32U controller board was used to motorize the mechanism by connecting the servo motors joined to each link's coupler to a corresponding port on the board, as shown in figure 8. This board is capable of controlling the speed of the servo motors, which was necessary for having a smooth motion and minimize vibrations generated during motion. Moreover, the SSC-32U board has a built-in motor driver, thus eliminating the need for additional motor shield or driver boards. This helps conserve space in the mechanism.

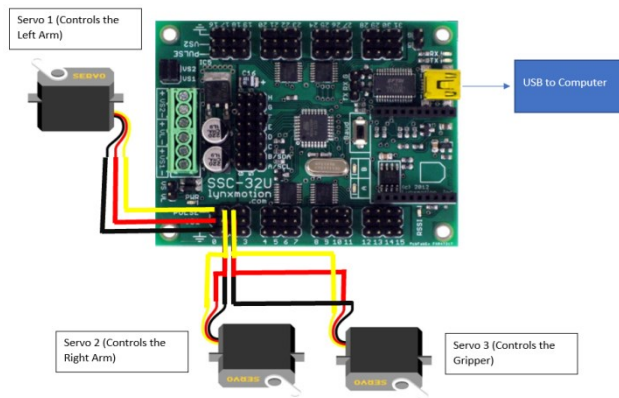


Fig. 8. SSC-32U board connected with the motors

An inertial measurement unit (IMU) was used to collect the real time data of the pitch angle for the platform while the mechanism is moving to its final desired position. The IMU unit (MPU 6050) measures acceleration in its own x, y, and z reference frames, which are relative to the inertial pose [8]. The flow chart shown in figure 9 summarizes the entire process, starting with the desired command from the user for angle θ_2 till the stage where the IMU takes readings of the platform pitch tilting angle. Figure 10 demonstrates the test bench used to validate the proposed approach. The manipulator was hanged freely in the center of $1m^3$ frame to simulate the caring drone free motion in 3D. The resulting tilting pitch angle of the attached mechanism relative to the quadrotor platform's real-time testing was collected while grasping a water bottle with a mass of 200 g, two cases were considered with and without the counterweight active balancing. As shown in the figure the test is done twice, once when the CoG compensation mechanism i.e. (counterweight) is absent (left) and once when it is present (right) respectively. The mechanism without the counterweight showed a tilting pitch angle of approximately 12° . After adding the counterweight, an improvement of 79% where the mechanism tilted only 2.5° . The testing platform's strings

caused these miniscule errors, which simulate the quadrotor hovering in real time. The autopilot system can stabilize these minor discrepancies, where there is no interaction with the implemented mechanism subsystem. Figure 11 presents the collected angle data from the IMU after the two experiments.

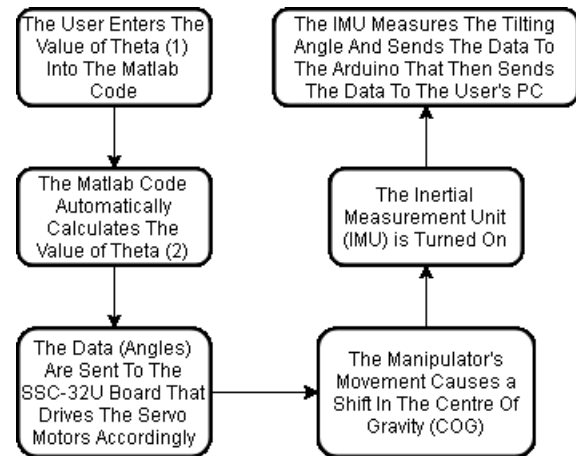


Fig. 9. Flow chart summarizes the COG shifting process compensation.

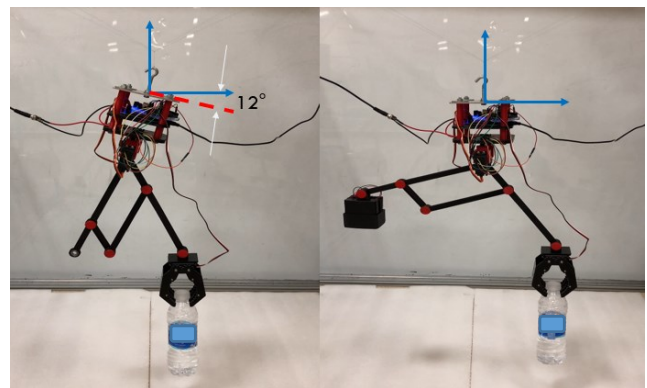


Fig. 10. Mechanism tilting angle without (left) and with (right) a counterweight mechanism.

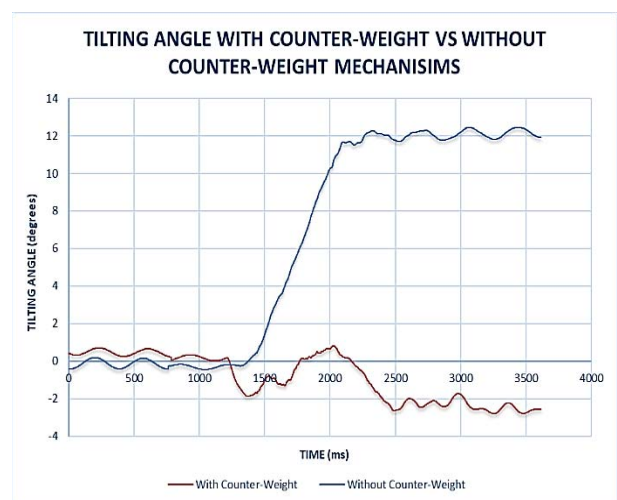


Fig. 11. Tilting pitch angle (degrees) vs time (ms) with and without a counterweight active mechanism

V. CONCLUSION

In this research a lightweight aerial manipulator was proposed to be deployed in quadrotors. The proposed design was able to rectify the possible CoG drift due to the manipulator motion or object influence. As it was demonstrated from the simulation and the experimental results, the proposed mechanism was capable to compensate for the CoG changes by an active counter balance weight inspired by pantograph-like structure. The final manipulator was able to increase the workspace addressed by the quadrotor compared to gripper-based manipulators attached to the quadrotors for pick and place applications, enhancing the stability for quadrotors with limited controller capabilities. However, the proposed design tried to compromise between the achieved benefits and possible limitations for adding mobile manipulator for quadrotors such as decreasing the compactness of the system, and reducing the overall system payload.

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